Cheat sheet

LLMOps & Deployment

Ship, evaluate, monitor and optimise

Stack

UI	Streamlit, Gradio, static HTML on GitHub Pages
Backend	FastAPI, serverless functions
Serving open models	vLLM, TGI, Ollama, llama.cpp
Packaging	Docker; pin dependencies (`requirements.txt`)
Free hosting	Hugging Face Spaces, Streamlit Community Cloud, GitHub Pages (static), Render free tier

Evaluation

Offline	Golden set, regression tests, LLM-as-judge, human review
Online	A/B tests, thumbs up/down, task completion, escalation rate
Gates	Block deploys if scores drop below thresholds

Observability

Traces	Prompt, context, tool calls, outputs, tokens, latency per step
Tools	Langfuse, Arize Phoenix, OpenTelemetry (open source options)
Alerts	Error spikes, latency p95, cost per day, safety flags

Optimisation

Cost	Model routing, caching, prompt compression, batching, shorter outputs
Latency	Streaming, smaller models, parallel retrieval, fewer chunks, regional deployment

Quality	Better retrieval, reranking, few-shot examples, fine-tuning

Versioning

Prompts	Stored in Git with IDs
Models	Pin exact model versions; test before upgrading
Data	Version indexes and datasets